



Chennai Mathematical Institute

MSc Data Science Programme

PageRank

Mathematical Foundations and Analysis

Authors

Anupam Dasgupta
MDS202509

Arkaprabha Chakraborty
MDS202510

Arnab Chakraborti
MDS202511

Arushi Marwaha
MDS202512

Supervisor

Professor Priyavrat C Deshpande
Chennai Mathematical Institute

pdeshpande@cmi.ac.in
Dr. Saipriya Dubey
Chennai Mathematical Institute

Contributions

Team Member	Contributions
Anupam Dasgupta	Python implementation via sparse matrices, Personalized PageRank, distribution analysis, node embeddings, and graph kernels, Extensions of PageRank: Kernels and Embeddings .
Arkaprabha Chakraborty	Mathematical motivation, modeling networks as graphs, adjacency matrices, and foundational properties of Markov chains.
Arnab Chakraborti	Google matrix formulation, damping factor, power iteration method, convergence rates, and computational efficiency, Empirical Results and Performance Analysis.
Arushi Marwaha	Random surfer interpretation, eigenvector formulation, and baseline model limitations including dangling nodes and rank sinks, Spectral Analysis of Reducible Markov Chains.

Abstract

This report analyzes the PageRank algorithm, covering its mathematical foundations, computational implementation, empirical behavior, and machine learning applications. PageRank formalizes web navigation as a discrete-time Markov chain. To guarantee convergence despite structural irregularities like rank sinks, a damping factor is introduced to construct the Google matrix. The algorithm is efficiently implemented using sparse matrix operations and the power iteration method, allowing computational complexity to scale linearly with the number of edges.

Empirical testing demonstrates that graph topology directly shapes PageRank distributions. Random graphs yield relatively uniform importance, whereas scale-free networks produce highly skewed distributions dominated by central hubs. Beyond simple ranking, PageRank functions as a structural feature extraction mechanism. By analyzing diffusion profiles and steady-state vectors, the algorithm enables the creation of rich node embeddings and graph kernels to quantitatively measure similarities between networks.

Contents

Contributions	2
1 Introduction	5
1.1 The PageRank Algorithm: Motivation	5
1.2 How it works	5
1.3 Elements of Linear Algebra Studied	6
2 Graph Representation	6
2.1 Modeling Web Pages as a Graph	6
2.2 The Adjacency Matrix	6
2.3 Example of a Graph and Adjacency Matrix	6

2.4	Arrangement of the Linear System	7
3	Transition Matrix and Markov Chains	7
3.1	Markov Chains Defined	7
3.2	Aperiodic and Irreducible Markov Chains	7
3.3	Necessity for PageRank Analysis	7
3.4	Converting the Adjacency Matrix to a Transition Matrix	7
3.5	Continuing the Example: Transition Matrix and Stationary Probabilities	8
4	Random Surfer Interpretation	8
5	Eigenvector Formulation	9
5.1	Spectral Properties	9
5.2	Component-wise Representation	9
5.3	Iterative Approximation	10
6	Limitations of the Basic Model	10
6.1	Dangling Nodes	10
6.2	Rank Sinks	10
6.3	Reducibility	11
6.3.1	Reducibility and Spectral Multiplicity	11
6.4	Periodicity	13
7	Google Matrix and Damping Factor	13
8	Linear System Formulation	14
9	Power Iteration Method	14
9.1	Convergence	15
9.2	Convergence Rate	15
10	Computational Efficiency	16
11	Empirical Results and Performance Analysis	16
11.1	Power Iteration Convergence	16
11.2	Distribution of PageRank Scores	17
11.3	Memory Comparison	17
11.4	Complexity Analysis	17
12	Extensions of PageRank: Kernels and Embeddings	18
12.1	Graph Kernels and PageRank	18
12.2	Node Embeddings from PageRank	19
12.3	Node-Level Kernels Based on Diffusion	19
12.4	PageRank as a Kernel Operator	20
12.5	Summary	20
13	Implementation and Results	20
13.1	Construction of the Transition Structure	21
13.2	Iterative Computation of PageRank	21
13.3	Graph Models Used	21

13.4 Application on a Random Graph	22
13.5 Application on a Scale-Free Graph	23
13.6 Application on the Karate Club Network	24
13.7 Personalized PageRank	24
13.8 Convergence Analysis	25
13.9 Graph Representations using PageRank	26
13.9.1 Node-Level Representation	26
13.9.2 Graph-Level Representation	26
13.10 Connection to Graph Kernels	27
13.10.1 Weisfeiler–Lehman Graph Kernel	27
13.10.2 Observations	29
13.11 Summary	29
13.12 Repository	30
References	31

Notation and Definitions

We define the key quantities used throughout the document:

- n : Number of nodes (webpages) in the graph.
- $\pi \in \mathbb{R}^n$: The PageRank vector, where π_i represents the importance (stationary probability) of node i , and:

$$\sum_{i=1}^n \pi_i = 1$$

- $\mathbf{1} \in \mathbb{R}^n$: The all-ones vector:

$$\mathbf{1} = [1, 1, \dots, 1]^T$$

- $P \in \mathbb{R}^{n \times n}$: Row-stochastic transition matrix derived from the link structure.
- $G \in \mathbb{R}^{n \times n}$: Google matrix incorporating damping[cite: 11].
- $\alpha \in (0, 1)$: Damping factor representing probability of following a link.
- d_i : Out-degree (number of outgoing links) of node i .
- $|E|$: Number of edges in the graph.

1 Introduction

1.1 The PageRank Algorithm: Motivation

PageRank is a computational algorithm designed to evaluate and assign a numerical weighting to nodes within a directed graph, specifically intended to measure relative importance . The primary motivation behind its creation was to improve information retrieval on the World Wide Web . Early search systems largely ranked pages based on content matching (e.g., keyword frequency), which was susceptible to manipulation . PageRank introduced a systemic approach relying on the structural links between pages, positing that a link from one page to another acts as an endorsement of relevance .

1.2 How it works

The algorithm works by formalizing the concept of importance recursively: a node is considered highly important if it receives incoming links from other highly important nodes . Mathematically, it works by modeling the network as a state space and finding a steady-state equilibrium . By translating the link structure into a system of linear equations, it computes a probability distribution that represents the likelihood of landing on any given node after traversing the network indefinitely .

1.3 Elements of Linear Algebra Studied

Analyzing PageRank requires several foundational concepts from linear algebra :

- **Matrices and Vectors:** For modeling the network structure and state probabilities .
- **Linear Systems:** To arrange and solve the simultaneous equations defining node importance .
- **Eigenvalues and Eigenvectors:** PageRank is fundamentally the dominant eigenvector of a modified transition matrix .
- **Stochastic Matrices:** Matrices with non-negative real entries where each row (or column) sums to 1, vital for conserving probability .

2 Graph Representation

2.1 Modeling Web Pages as a Graph

Web pages and their hyperlinks are naturally modeled as a directed graph because the system consists of distinct entities connected by directional relationships . In this formulation, each webpage is a node (or vertex), and each hyperlink from page i to page j is a directed edge . This abstraction strips away the content of the pages, leaving only the structural relationships needed to analyze connectivity .

2.2 The Adjacency Matrix

A directed graph is mathematically represented using an adjacency matrix, denoted as $A \in \mathbb{R}^{n \times n}$. The elements of this matrix are defined as:

$$A_{ij} = \begin{cases} 1 & \text{if there is a link from node } i \text{ to node } j \\ 0 & \text{otherwise} \end{cases}$$

2.3 Example of a Graph and Adjacency Matrix

Consider a simple graph with $n = 3$ nodes labeled 1, 2, and 3 .

- Node 1 links to Node 2 and Node 3 .
- Node 2 links only to Node 3 .
- Node 3 links only to Node 1 .

The out-degrees are $d_1 = 2$, $d_2 = 1$, and $d_3 = 1$. The adjacency matrix A for this graph is:

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

2.4 Arrangement of the Linear System

The core principle of PageRank is that the importance π_j of a node j is the sum of the importance scores of all nodes i linking to it, divided by their respective out-degrees d_i . This is formulated as a linear system:

$$\pi_j = \sum_{i=1}^n A_{ij} \frac{\pi_i}{d_i}$$

By writing this out for every node in the graph, we create a system of n linear equations that must be solved simultaneously to find the vector π .

3 Transition Matrix and Markov Chains

3.1 Markov Chains Defined

A Markov chain is a mathematical system that undergoes transitions from one state to another within a finite state space. Its defining characteristic is the "Markov property" or memorylessness: the probability of transitioning to any future state depends solely on the current state and not on the sequence of events that preceded it.

3.2 Aperiodic and Irreducible Markov Chains

- **Aperiodic Markov Chain:** A chain is aperiodic if the system cannot get trapped in deterministic cycles (where return times to a state are strict multiples of an integer greater than 1).
- **Irreducible Markov Chain:** A chain is irreducible if it is possible to get to any state from any other state; there are no isolated components or terminal absorbing states.

3.3 Necessity for PageRank Analysis

These properties are strictly required due to the Perron-Frobenius theorem. If a Markov chain is both irreducible and aperiodic, it is guaranteed to possess a unique, positive stationary probability distribution. Furthermore, it guarantees that standard iterative methods (like the power method) will converge to this distribution regardless of the initial starting state. Without these properties, the system of equations might have multiple solutions or none at all.

3.4 Converting the Adjacency Matrix to a Transition Matrix

To model the system as a Markov chain, the adjacency matrix A must be converted into a row-stochastic transition matrix P . This is achieved by dividing each element by the out-degree d_i of its corresponding row i :

$$P_{ij} = \frac{A_{ij}}{d_i}$$

Each row of P now represents the transition probabilities from a given node, and the sum of probabilities in every row equals 1.

3.5 Continuing the Example: Transition Matrix and Stationary Probabilities

Using our previous 3-node graph with $d_1 = 2, d_2 = 1, d_3 = 1$, the transition matrix P is:

$$P = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

To find the stationary probabilities $\pi = [\pi_1, \pi_2, \pi_3]^T$, we solve the equation $\pi^T = \pi^T P$, subject to $\sum_{i=1}^n \pi_i = 1$:

$$\begin{aligned} \pi_1 &= \pi_3(1) \\ \pi_2 &= \pi_1(1/2) \\ \pi_3 &= \pi_1(1/2) + \pi_2(1) \end{aligned}$$

Substituting $\pi_2 = 0.5\pi_1$ into the third equation yields $\pi_3 = 0.5\pi_1 + 0.5\pi_1 = \pi_1$. Thus, we have $\pi_1 = \pi_3$ and $\pi_2 = 0.5\pi_1$. Using the sum constraint $\pi_1 + \pi_2 + \pi_3 = 1$:

$$\pi_1 + 0.5\pi_1 + \pi_1 = 1 \implies 2.5\pi_1 = 1 \implies \pi_1 = 0.4$$

Consequently, the stationary probabilities (PageRank scores) are :

$$\begin{aligned} \pi_1 &= 0.4 \\ \pi_2 &= 0.2 \\ \pi_3 &= 0.4 \end{aligned}$$

This simple example yields an exact stationary distribution because the base graph is inherently irreducible and aperiodic .

4 Random Surfer Interpretation

The PageRank formulation can be rigorously modeled as a discrete-time Markov chain over a directed graph of webpages . Let $P \in \mathbb{R}^{n \times n}$ denote the row-stochastic transition matrix derived from the link structure, and let $x^{(k)} \in \mathbb{R}^n$ represent the probability distribution of a surfer's location after k steps, satisfying :

$$x_i^{(k)} \geq 0, \quad \sum_{i=1}^n x_i^{(k)} = 1.$$

The evolution of the system is governed by :

$$x^{(k+1)} = P^T x^{(k)}. \tag{1}$$

This defines a finite Markov chain with state space corresponding to webpages, where transition probabilities are determined by normalized outgoing links . The transpose appears due to the convention of representing probability distributions as column vectors .

Under this formulation, the PageRank vector π is defined as a stationary distribution of the Markov chain, i.e., a vector satisfying :

$$P^T \pi = \pi, \quad \pi_i \geq 0, \quad \sum_{i=1}^n \pi_i = 1. \quad (2)$$

If the Markov chain is irreducible and aperiodic, standard results guarantee the existence of a unique stationary distribution, and for any initial distribution $x^{(0)}$, the sequence $\{x^{(k)}\}$ converges :

$$\lim_{k \rightarrow \infty} x^{(k)} = \pi. \quad (3)$$

Thus, PageRank assigns to each node the limiting probability of visiting that node under repeated random navigation of the graph .

5 Eigenvector Formulation

The stationary condition

$$P^T \pi = \pi \quad (4)$$

implies that π is a right eigenvector of P^T associated with the eigenvalue $\lambda = 1$. Equivalently, π lies in the null space of $(I - P^T)$:

$$(I - P^T)\pi = 0. \quad (5)$$

5.1 Spectral Properties

Since P is row-stochastic, its transpose P^T is column-stochastic . Therefore:

- $\lambda = 1$ is always an eigenvalue of P^T
- All eigenvalues satisfy $|\lambda_i| \leq 1$
- The spectral radius $\rho(P^T) = 1$

If P^T is primitive (i.e., irreducible and aperiodic), then by the Perron-Frobenius theorem :

- The eigenvalue $\lambda = 1$ is simple (algebraic multiplicity 1)
- The corresponding eigenvector π is strictly positive
- π is unique up to scaling, and normalization enforces $\sum_i \pi_i = 1$

5.2 Component-wise Representation

The eigenvector equation can be expressed entry-wise as :

$$\pi_i = \sum_{j=1}^n P_{ji} \pi_j = \sum_{j \in \mathcal{B}_i} \frac{\pi_j}{d_j}. \quad (6)$$

This relation shows that the rank of a node is a weighted sum of the ranks of nodes linking to it, with weights inversely proportional to their out-degrees .

5.3 Iterative Approximation

The stationary distribution can be approximated using the power iteration :

$$\pi^{(k+1)} = P^T \pi^{(k)}. \quad (7)$$

Expanding recursively,

$$\pi^{(k)} = (P^T)^k \pi^{(0)},$$

which converges to π provided the Markov chain satisfies the conditions of ergodicity . The rate of convergence is governed by the magnitude of the second-largest eigenvalue of P^T .

6 Limitations of the Basic Model

Although the formulation $\pi = P^T \pi$ is mathematically well-defined, it fails to produce a reliable ranking in general web graphs due to violations of key Markov chain assumptions .

6.1 Dangling Nodes

A dangling node is a node with no outgoing links ($d_i = 0$) . In this case, the corresponding row of P is identically zero, which implies :

$$\sum_{j=1}^n P_{ij} = 0 \neq 1.$$

Thus, P ceases to be stochastic, and probability mass is not conserved :

$$\|x^{(k+1)}\|_1 < \|x^{(k)}\|_1.$$

This leads to degeneration of the Markov process and prevents convergence to a valid stationary distribution .

6.2 Rank Sinks

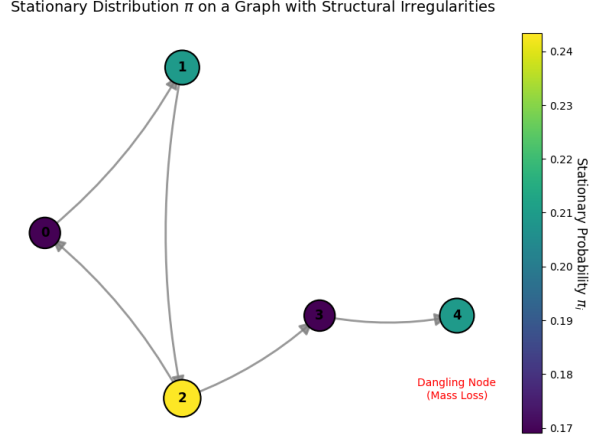
A rank sink is a subset of nodes $S \subseteq \{1, \dots, n\}$ such that :

$$\forall i \in S, \quad \text{all outgoing links from } i \text{ remain within } S.$$

This induces a closed communicating class in the Markov chain . Once the process enters S , it cannot leave, implying that :

$$\lim_{k \rightarrow \infty} \sum_{i \in S} x_i^{(k)} = 1.$$

Consequently, all probability mass concentrates within the sink, and nodes outside S receive zero rank asymptotically .



As illustrated in Figure , a rank sink occurs when a cluster of nodes S has no outgoing edges to the rest of the graph $V \setminus S$. Mathematically, this implies that for all $i \in S$ and $j \notin S$, the transition probability $P_{ij} = 0$.

In the visualized model, nodes $\{0, 1, 2\}$ form a closed communicating class. Because the random surfer cannot escape this subgraph once they enter, the power iteration $\pi^{(k+1)} = P^T \pi^{(k)}$ results in all probability mass concentrating within S as $k \rightarrow \infty$. Consequently:

$$\sum_{i \in S} \pi_i = 1, \quad \text{while} \quad \pi_j = 0 \text{ for all } j \notin S. \quad (8)$$

This creates a black hole effect where the global importance of the web graph is ignored in favor of a local loop, necessitating the use of a damping factor to restore ergodicity.

6.3 Reducibility

If the graph is not strongly connected, then P^T is reducible and can be brought to block upper-triangular form via permutation :

$$P^T = \begin{bmatrix} P_1 & * & \cdots & * \\ 0 & P_2 & \cdots & * \\ \vdots & \vdots & \ddots & * \\ 0 & 0 & \cdots & P_k \end{bmatrix}.$$

In such cases :

- Multiple invariant subspaces exist
- The eigenvalue $\lambda = 1$ may have multiplicity greater than one
- The stationary distribution is not unique

6.3.1 Reducibility and Spectral Multiplicity

A directed graph $\mathcal{G} = (V, E)$ is reducible if it is not strongly connected, implying the existence of at least two disjoint sets of nodes $S_1, S_2 \subset V$ such that no path exists from S_1 to S_2 . In such instances, the transition matrix P^T (after a symmetric permutation of rows and columns) assumes a block diagonal or block upper-triangular form:

$$P^T = \begin{bmatrix} P_1 & \mathbf{0} \\ \mathbf{0} & P_2 \end{bmatrix} \quad (9)$$

where $P_1 \in \mathbb{R}^{n_1 \times n_1}$ and $P_2 \in \mathbb{R}^{n_2 \times n_2}$ represent the local transition dynamics within their respective closed communicating classes.

Mathematical Implications

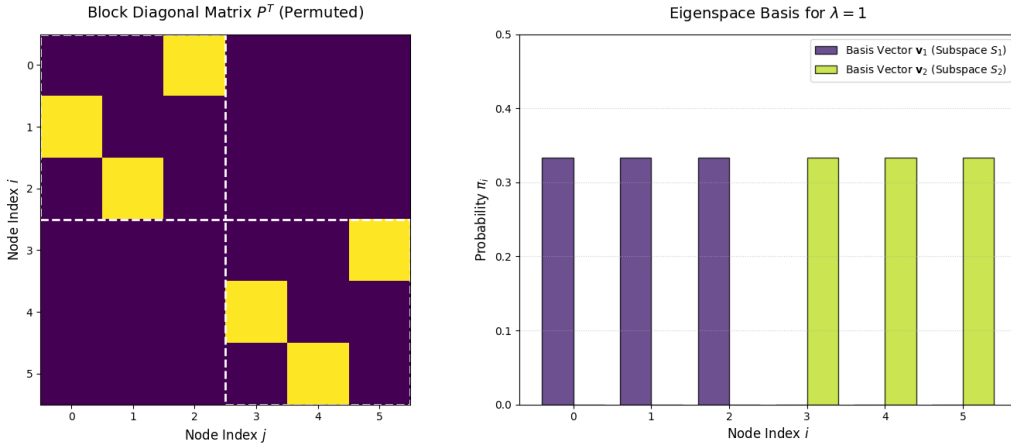
The structural reducibility leads to several critical failures in the standard PageRank formulation:

1. **Geometric Multiplicity of $\lambda = 1$:** Since P_1 and P_2 are themselves stochastic matrices, the Perron-Frobenius theorem guarantees that each sub-matrix has a dominant eigenvalue $\lambda_i = 1$. Consequently, the global matrix P^T possesses an eigenvalue $\lambda = 1$ with an algebraic multiplicity $k \geq 2$.
2. **Non-Uniqueness of π :** The eigenspace $E(1)$ corresponding to $\lambda = 1$ is at least two-dimensional. If $\mathbf{v}_1$ is the stationary distribution for P_1 and $\mathbf{v}_2$ for P_2 , then any convex combination:

$$\pi = \beta \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{0} \end{bmatrix} + (1 - \beta) \begin{bmatrix} \mathbf{0} \\ \mathbf{v}_2 \end{bmatrix}, \quad \beta \in [0, 1] \quad (10)$$

constitutes a valid stationary distribution.

3. **Dependency on Initial Conditions:** Under power iteration, the sequence $\{x^{(k)}\}$ fails to converge to a global rank. Instead, the final distribution becomes a function of the initial probability mass distribution $x^{(0)}$ across the invariant subspaces.



Spectral decomposition of a reducible Markov chain. The block-diagonal structure of P^T (left) corresponds to the disjoint invariant subspaces S_1 and S_2 , resulting in a non-unique eigenspace basis for $\lambda = 1$ (right). The computational logic and spectral plots can be found in the Spectral Analysis Repository.

6.4 Periodicity

If the Markov chain is periodic with period $d > 1$, then the sequence $\{x^{(k)}\}$ does not converge but instead oscillates :

$$x^{(k+d)} = x^{(k)}.$$

This prevents the existence of a stable limiting distribution .

Conclusion of Limitations

These issues demonstrate that the basic PageRank formulation does not guarantee :

- existence of a unique stationary distribution,
- convergence of iterative methods,
- robustness to structural irregularities in the graph.

To overcome these limitations, the transition matrix must be modified to enforce irreducibility and aperiodicity, leading to the construction of the Google matrix with damping .

7 Google Matrix and Damping Factor

The PageRank algorithm, introduced by *Google*, models the web as a directed graph and assigns importance scores to webpages based on link structure . The algorithm relies on a Markov chain interpretation of web navigation and computes a stationary distribution over nodes .

Let P be the row-stochastic transition matrix defined as :

$$P_{ij} = \begin{cases} \frac{1}{d_i}, & \text{if page } i \text{ links to } j \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

where d_i is the out-degree of node i .

Real web graphs contain dangling nodes and rank sinks, which lead to reducibility . To address this, we introduce a damping factor $\alpha \in (0, 1)$ and define the Google matrix :

$$G = \alpha P + (1 - \alpha)E \quad (12)$$

where :

$$E = \frac{1}{n} \mathbf{1}\mathbf{1}^T \quad (13)$$

Here, $\mathbf{1}\mathbf{1}^T$ denotes an $n \times n$ matrix where every entry is equal to 1 .

The matrix G ensures :

- Irreducibility
- Aperiodicity
- Existence of a unique stationary distribution

Typically, $\alpha = 0.85$, balancing link-following behavior and random jumps .

8 Linear System Formulation

The PageRank vector π satisfies:

Here, π represents the stationary distribution of the Markov chain, where π_i denotes the long-run probability of visiting node i .

$$\pi^T G = \pi^T \quad (14)$$

or equivalently :

$$G^T \pi = \pi \quad (15)$$

Substituting G :

$$\pi = \alpha P^T \pi + (1 - \alpha) \frac{1}{n} \mathbf{1} \quad (16)$$

Rewriting :

$$(I - \alpha P^T) \pi = \frac{1 - \alpha}{n} \mathbf{1} \quad (17)$$

This defines a linear system :

$$A \pi = b \quad (18)$$

where :

$$A = I - \alpha P^T \quad (19)$$

$$b = \frac{1 - \alpha}{n} \mathbf{1} \quad (20)$$

The matrix A is non-singular, ensuring a unique solution .

9 Power Iteration Method

Due to the scale of web graphs, direct solutions are infeasible . Instead, the PageRank vector is computed using the power iteration method :

$$\pi^{(k+1)} = G^T \pi^{(k)} \quad (21)$$

or :

$$\pi^{(k+1)} = \alpha P^T \pi^{(k)} + (1 - \alpha) \frac{1}{n} \mathbf{1} \quad (22)$$

Starting with :

$$\pi^{(0)} = \frac{1}{n} \mathbf{1} \quad (23)$$

the iteration continues until convergence .

Intuitively, each iteration represents one step of a random surfer navigating the web . With probability α , the surfer follows a hyperlink, and with probability $1 - \alpha$, jumps

uniformly at random to any page . The vector $\pi^{(k)}$ therefore represents the probability distribution of the surfer's location after k steps .

As iterations proceed, probability mass flows through the graph along outgoing links and is redistributed across nodes . This process gradually stabilizes, and the distribution converges to a steady state . In practice, the iteration is terminated when successive updates differ by a small amount, typically measured using a norm :

$$\|\pi^{(k+1)} - \pi^{(k)}\| < \epsilon \quad (24)$$

for some small tolerance ϵ .

9.1 Convergence

The matrix G satisfies the conditions of the Perron-Frobenius theorem :

- G is stochastic, meaning its rows sum to 1
- The teleportation term makes every entry of G positive
- The corresponding Markov chain is irreducible and aperiodic

These properties guarantee that the dominant eigenvalue of G is 1 and that there exists a unique stationary distribution π . Thus:

$$\pi^{(k)} \rightarrow \pi \quad (25)$$

Intuitively, convergence occurs because each iteration redistributes probability mass across the graph while preventing probability from getting trapped in dead ends, disconnected components, or cyclic rank sinks . The damping term ensures that at every step, a small amount of probability is spread uniformly across all nodes . Therefore, repeated multiplication by G^T drives any initial distribution toward the same unique stationary distribution .

9.2 Convergence Rate

The convergence rate depends on the spectral gap :

$$1 - |\lambda_2| \quad (26)$$

where λ_2 is the eigenvalue of G with the second largest magnitude . A larger spectral gap implies faster convergence, because the components of the initial vector corresponding to non-dominant eigenvalues decay more quickly during power iteration .

Teleportation ensures :

$$|\lambda_2| \leq \alpha \quad (27)$$

Thus, the damping factor directly controls the worst-case convergence rate . A smaller value of α generally increases the spectral gap and speeds up convergence, while a larger value of α places more emphasis on the original link structure but may require more iterations to converge .

10 Computational Efficiency

Each iteration requires sparse matrix-vector multiplication, with complexity :

$$O(|E|) \tag{28}$$

where $|E|$ is the number of edges . This allows PageRank to scale to very large graphs

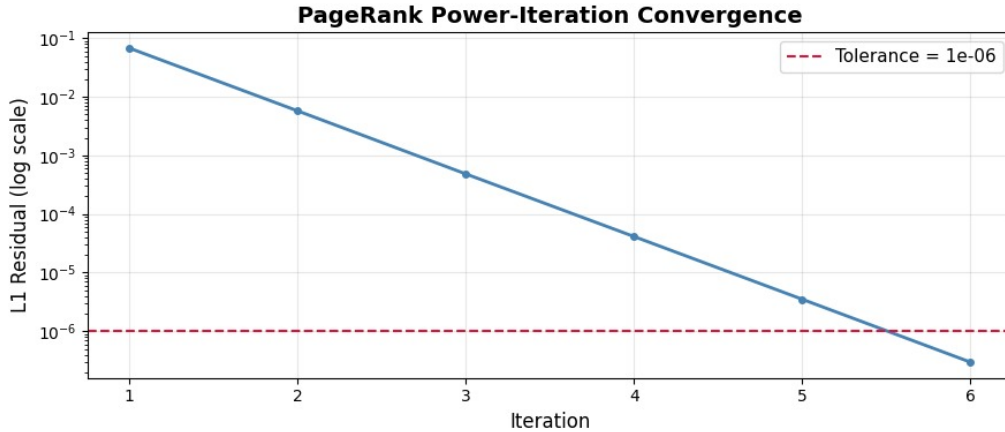
The sparse matrix representation is effective because real web graphs are extremely sparse: each webpage links to only a small fraction of all webpages . Therefore, most entries of the transition matrix P are zero and do not need to be stored explicitly . Instead of storing the full $n \times n$ matrix, it is sufficient to store only the non-zero entries corresponding to actual hyperlinks .

In practice, the update can be computed by distributing the current PageRank score of each node across its outgoing edges . Thus, each iteration scans only the existing links in the graph rather than all possible pairs of webpages . This reduces the cost from $O(n^2)$ for dense matrix multiplication to $O(|E|)$ for sparse graph traversal, making the power iteration method feasible for large-scale web graphs .

documentclass[11pt]article

11 Empirical Results and Performance Analysis

11.1 Power Iteration Convergence



Asymptotic convergence of the PageRank vector $\pi^{(k)}$. The plot demonstrates the L_1 error reduction over successive iterations. The complete source code and visualization logic can be accessed at the [Repository](#).

The convergence plot shows the decay of the L_1 residual,

$$\|\pi^{(k+1)} - \pi^{(k)}\|_1,$$

over successive iterations. The residual decreases rapidly on a logarithmic scale and falls below the tolerance level of 10^{-6} by the sixth iteration. This indicates that the PageRank vector stabilizes quickly for the generated sparse graph.

The fast convergence is consistent with the damping-based PageRank formulation. The damping factor prevents probability mass from being trapped in disconnected regions or rank sinks, while the teleportation term ensures that the Markov chain mixes efficiently.

11.2 Distribution of PageRank Scores

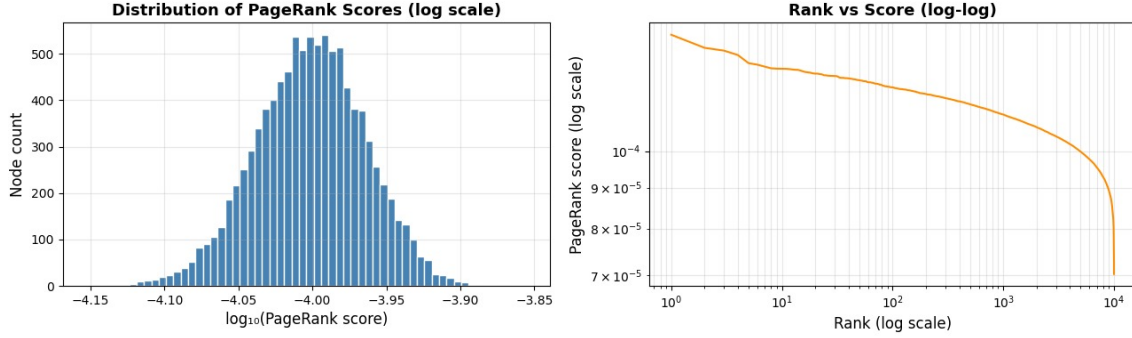


Figure 1: Distribution of PageRank scores and rank-score behavior.

The left plot shows the distribution of $\log_{10}$ PageRank scores across all nodes. Most nodes have PageRank values concentrated around similar magnitudes, indicating that importance is spread across the sparse random graph rather than being dominated by only a few nodes.

The right plot shows the rank versus PageRank score on a log-log scale. Higher-ranked nodes have larger PageRank values, while the scores gradually decline as rank increases. The curve shows that PageRank still differentiates node importance, even though the graph is randomly generated and relatively uniform.

11.3 Memory Comparison

The implementation compares dense matrix storage with sparse CSR matrix storage.

$$\text{Dense matrix memory} = 762.9 \text{ MB}$$

$$\text{CSR sparse matrix memory} = 11.4 \text{ MB}$$

$$\text{Compression ratio} = 66.8\times$$

This shows why sparse representation is essential for PageRank. A dense matrix stores all n^2 possible entries, even though most of them are zero. In contrast, CSR storage keeps only the non-zero entries corresponding to actual edges. This reduces memory usage significantly and makes computation feasible for large graphs.

11.4 Complexity Analysis

The graph contains:

$$|E| = 994,828$$

edges after removing self-loops and duplicate edges.
The space complexity is:

$$O(|E|) = O(994,828)$$

because only existing edges are stored.

The time complexity of power iteration is:

$$O(k|E|)$$

where k is the number of iterations required for convergence. Since convergence occurred in 6 iterations:

$$\begin{aligned} O(k|E|) &= O(6 \times 994,828) \\ &= O(5,968,968) \end{aligned}$$

Thus, the algorithm scales linearly with the number of edges per iteration. This is much more efficient than dense matrix multiplication, which would require $O(n^2)$ operations per iteration.

12 Extensions of PageRank: Kernels and Embeddings

The PageRank algorithm, originally formulated as a stationary distribution over a Markov chain, provides a powerful foundation for understanding graph structure. Beyond ranking nodes, its underlying random walk interpretation enables extensions into graph kernels and node embeddings, which are central to modern graph-based machine learning.

From a machine learning perspective, PageRank can be viewed as a *feature extraction mechanism* driven entirely by the geometry of the graph. Rather than relying on node attributes, it constructs representations through repeated application of a linear operator, allowing structural information to propagate across the network.

12.1 Graph Kernels and PageRank

Definition. A graph kernel is a function

$$K : \mathcal{G} \times \mathcal{G} \rightarrow \mathbb{R}$$

that measures similarity between graphs by implicitly mapping them into a (possibly high-dimensional) feature space.

Traditional graph kernels are based on counting substructures such as walks, paths, or subtrees. In contrast, PageRank induces a global notion of similarity through diffusion over the graph. This shift replaces explicit enumeration of structures with an operator-driven aggregation process.

Let G_1 and G_2 be two graphs with corresponding PageRank vectors $\pi^{(1)}$ and $\pi^{(2)}$. A PageRank-based graph kernel can be defined as:

$$K(G_1, G_2) = \langle \pi^{(1)}, \pi^{(2)} \rangle.$$

This formulation treats PageRank as a feature representation of the graph. Since π is the fixed point of a linear operator, this kernel compares graphs in terms of their *equilibrium distributions*. In effect, it measures similarity not at the level of edges or local neighborhoods, but at the level of global steady-state behavior.

12.2 Node Embeddings from PageRank

Definition. A node embedding is a mapping

$$\phi : V \rightarrow \mathbb{R}^d$$

that assigns each node a vector representation.

From a linear algebraic perspective, the PageRank vector $\pi \in \mathbb{R}^n$ itself can be interpreted as a one-dimensional embedding:

$$\phi(i) = \pi_i.$$

However, this representation is inherently compressed: it reflects only the final equilibrium of a diffusion process. A richer embedding emerges when we consider the entire family of Personalized PageRank (PPR) vectors.

Definition. For a node s , the personalized PageRank vector $\pi^{(s)}$ satisfies:

$$\pi^{(s)} = \alpha P^T \pi^{(s)} + (1 - \alpha) e_s,$$

where e_s is the indicator vector for node s .

Using PPR, each node i can be embedded as:

$$\phi(i) = [\pi_i^{(1)}, \pi_i^{(2)}, \dots, \pi_i^{(n)}].$$

This construction can be interpreted as assigning to each node a coordinate system defined by *diffusion responses*. Each dimension corresponds to the influence of a source node, and the embedding captures how information flows through the graph toward node i .

From an operator viewpoint, this embedding encodes columns of the inverse operator $(I - \alpha P^T)^{-1}$, revealing how the graph transforms localized signals into global patterns. In practice, these high-dimensional embeddings are approximated or compressed, but the underlying principle remains the same: node representations arise from repeated applications of the transition operator.

12.3 Node-Level Kernels Based on Diffusion

Given node embeddings $\phi(i), \phi(j) \in \mathbb{R}^d$, a node-level kernel is defined as:

$$K(i, j) = \langle \phi(i), \phi(j) \rangle.$$

If embeddings are derived from Personalized PageRank, this becomes:

$$K(i, j) = \langle \pi^{(i)}, \pi^{(j)} \rangle.$$

This kernel compares nodes through their diffusion profiles. Two nodes are considered similar if they distribute influence across the graph in a similar way. Importantly, this notion of similarity is not tied to proximity in the graph, but rather to *functional similarity* under the action of the transition operator.

From a spectral perspective, this kernel implicitly emphasizes components aligned with dominant eigenvectors of P^T . As a result, it filters out high-frequency variations (localized noise) while preserving global structural patterns.

12.4 PageRank as a Kernel Operator

From the linear system formulation of PageRank:

$$(I - \alpha P^T)\pi = \frac{1 - \alpha}{n}\mathbf{1},$$

we obtain:

$$\pi = (I - \alpha P^T)^{-1}b, \quad \text{where } b = \frac{1 - \alpha}{n}\mathbf{1}.$$

Observation. The matrix

$$K = (I - \alpha P^T)^{-1}$$

acts as a kernel operator.

Expanding as a Neumann series:

$$K = \sum_{k=0}^{\infty} \alpha^k (P^T)^k,$$

which shows that PageRank aggregates contributions from walks of all lengths, with longer walks exponentially down-weighted.

This expansion provides a clear operator-theoretic interpretation: each power $(P^T)^k$ propagates information across paths of length k , while the factor α^k controls the decay of influence. Thus, the inverse operator acts as a *resolvent*, accumulating multi-scale structural information.

From a machine learning standpoint, this is precisely the behavior expected from a kernel: it maps inputs (nodes or graphs) into a space where similarity reflects accumulated interactions over multiple scales. Unlike explicit feature maps, this representation is implicit, encoded entirely in the linear operator.

12.5 Summary

PageRank provides a unifying framework linking graph theory, linear algebra, and machine learning. As a stationary distribution of a Markov chain, it captures global structure through diffusion. Graph kernels extend this idea to measure similarity between graphs, while node embeddings generalize it into vector representations suitable for learning tasks.

At its core, PageRank is the repeated application of a linear operator followed by normalization. Modern graph learning methods can be viewed as extensions of this principle, where the transition operator is used to construct increasingly expressive representations of graph structure.

13 Implementation and Results

In this section, we describe the implementation of the PageRank algorithm and analyze its behavior on different graph structures. The implementation was carried out in Python using sparse matrix operations.

13.1 Construction of the Transition Structure

Given an input graph in the form of an edge list, we first constructed the transition matrix implicitly. Instead of forming a dense matrix, we stored the structure using a sparse representation.

For each node i , its out-degree d_i was computed. The contribution of node i to its neighbors was then normalized by assigning weight $1/d_i$ to each outgoing edge. Using this, a sparse matrix $M = P^T$ was constructed in CSR format.

This representation ensured that:

- Only non-zero entries corresponding to edges were stored,
- Matrix-vector multiplication could be performed efficiently,
- Memory usage remained low even for moderately sized graphs.

13.2 Iterative Computation of PageRank

The PageRank vector was initialized as a uniform distribution over all nodes. The update was performed iteratively using sparse matrix-vector multiplication.

At each iteration:

- The term $M\pi^{(k)}$ was computed using sparse multiplication,
- The contribution from dangling nodes was calculated separately and redistributed,
- A teleportation term was added to ensure stability.

The process was repeated until convergence, which was measured using the L^1 -norm difference between successive iterates.

13.3 Graph Models Used

Erdős–Rényi (ER) Graph. An ER graph $G(n, p)$ is generated by connecting each pair of nodes independently with probability p . It has no built-in hierarchy or hubs, and node degrees concentrate around their mean. We use it as a baseline with minimal structural bias.

Barabási–Albert (BA) / Preferential Attachment Graph. A BA-type graph is constructed by sequentially adding nodes, where each new node connects to existing nodes with probability proportional to their current degree (preferential attachment). This yields a scale-free network with a heavy-tailed degree distribution and the emergence of hubs. In our implementation, we use a directed preferential-attachment variant to make it compatible with PageRank.

Karate Club Graph. The Karate Club graph is a real-world social network of interactions between members of a club that eventually split into two factions. It is commonly used as a benchmark because it contains meaningful community structure. We use it to verify that PageRank identifies central individuals in real data.

13.4 Application on a Random Graph

To study the behavior of PageRank in the absence of strong structural patterns, we applied the algorithm to an Erdős–Rényi random graph with fixed edge probability.

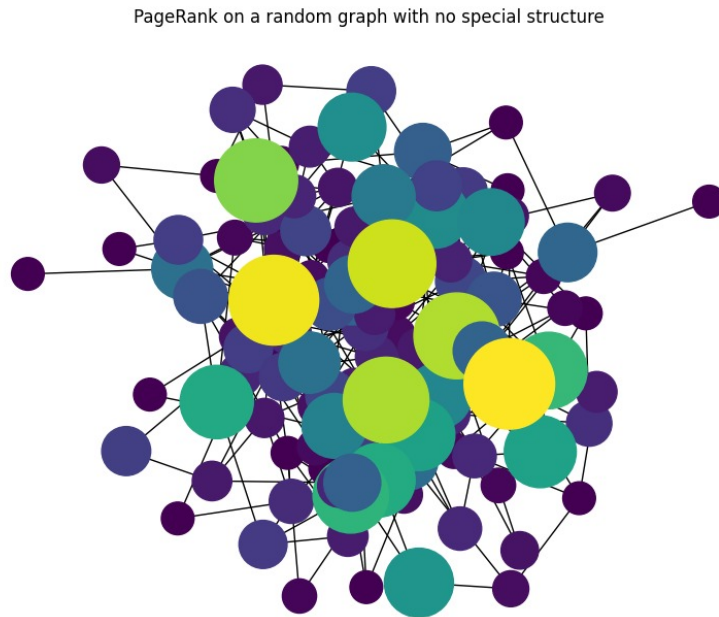


Figure 2: PageRank on a random graph with no special structure

The visualization shows that, unlike the scale-free graph, there are no strongly dominant nodes. While some nodes appear slightly larger, the differences in size and color are much less pronounced. This indicates that importance is relatively more evenly distributed across the network.

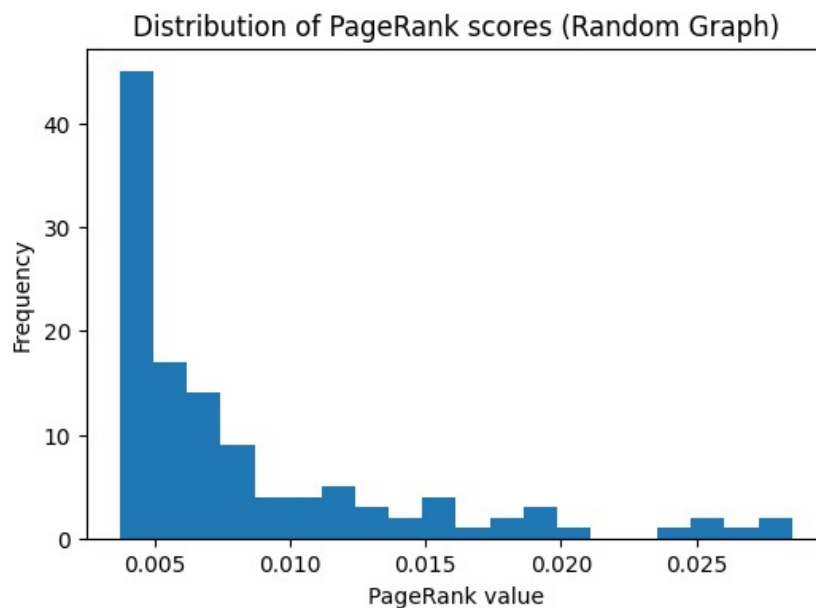


Figure 3: Histogram of the pagerank scores

The histogram further confirms this observation. Most nodes have PageRank values concentrated within a narrow range, with only a few nodes deviating slightly. There is no sharp separation between highly important and unimportant nodes.

From this, we observe that:

- PageRank does not produce strongly dominant nodes when the graph lacks structural bias.
- Importance is distributed more uniformly compared to scale-free networks.
- The output of PageRank is directly influenced by the underlying connectivity pattern of the graph.

This experiment highlights that PageRank reflects the structure of the graph: in random networks, it yields relatively uniform importance, while in structured networks, it leads to highly skewed distributions.

13.5 Application on a Scale-Free Graph

We applied the implementation to a Barabási–Albert graph with 120 nodes. This model is relevant because it captures the hub-dominated structure observed in many real-world networks such as the web.

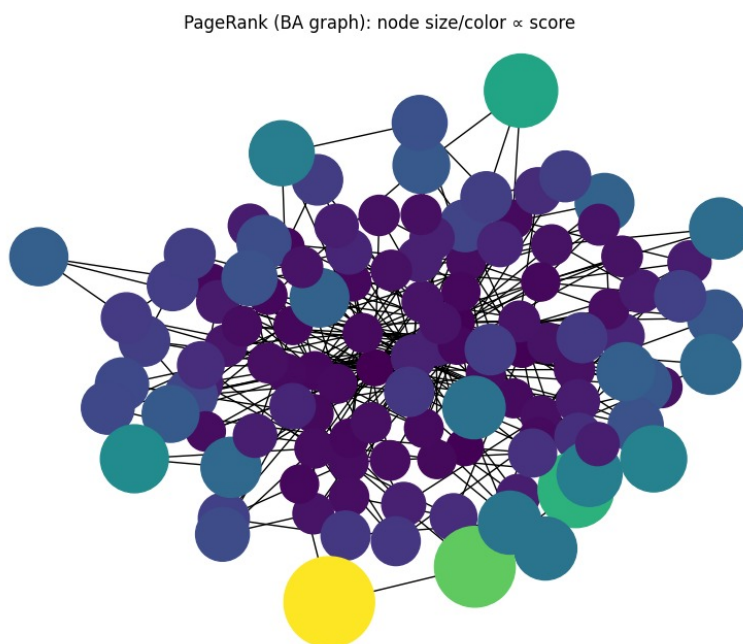


Figure 4: PageRank on a Barabási–Albert graph

The resulting PageRank distribution was highly skewed. A small number of nodes accumulated significantly higher scores, while the majority of nodes had very low values.

This indicates that importance is concentrated in a few regions of the graph. Nodes that receive connections from already important nodes gain disproportionately high scores, leading to the emergence of dominant hubs.

13.6 Application on the Karate Club Network

We further applied the algorithm to the Karate Club graph, which represents real-world social interactions. This allows us to test whether PageRank recovers meaningful influence patterns in real-world social networks.

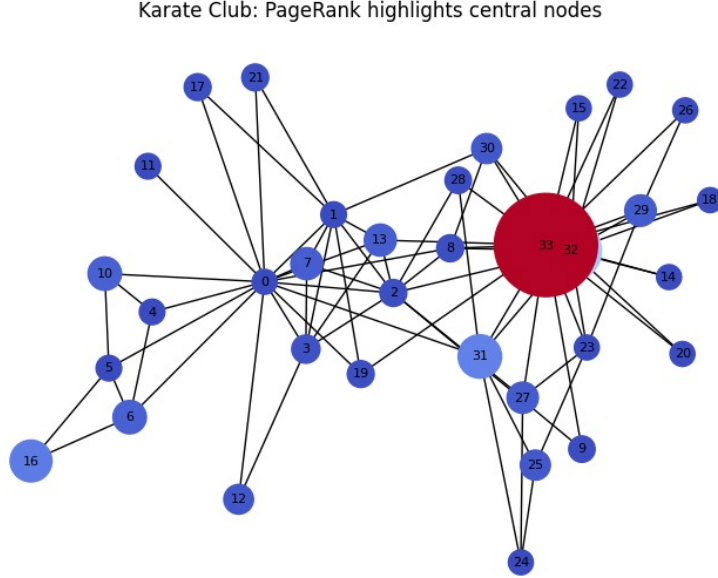


Figure 5: PageRank on the Karate Club graph

The results show that PageRank identifies a small number of central individuals as significantly more important than others. In particular, nodes corresponding to the leaders of the two factions receive the highest scores.

This demonstrates that the algorithm is able to recover meaningful structure from real-world data, highlighting individuals who play a central role in the connectivity of the network.

13.7 Personalized PageRank

Personalized PageRank (PPR) modifies the teleportation distribution to concentrate on a specific node s . The update becomes:

$$\pi^{(k+1)} = \alpha M \pi^{(k)} + (1 - \alpha) e_s,$$

where e_s is the indicator vector for node s .

This biases the random walk toward the source node, producing a localized importance distribution. In contrast to global PageRank, PPR captures node-specific neighborhoods and diffusion patterns. To analyze localized influence, we implemented Personalized PageRank by modifying the teleportation vector to concentrate on a single node.

PPR centered at node 0: localized influence

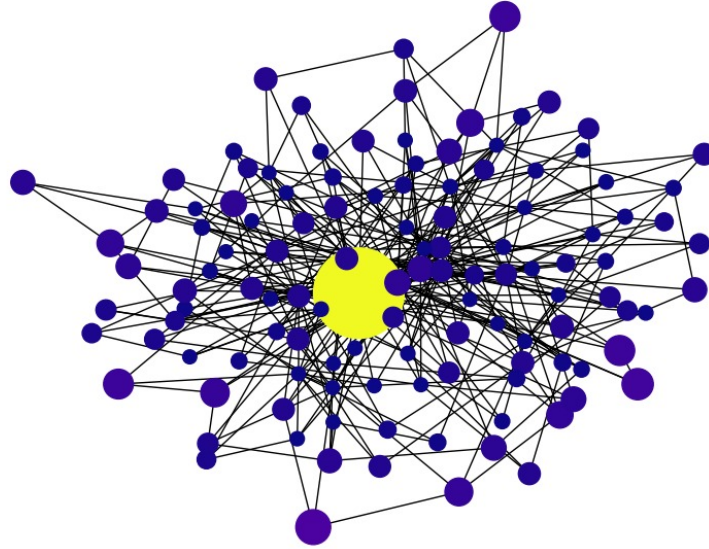


Figure 6: Personalised PageRank

The resulting distribution shows that importance is centered around the chosen node. Nodes directly connected to the source receive higher scores, while nodes further away receive negligible values.

This allows us to isolate the portion of the graph that is most relevant to a given node and analyze influence from a local perspective.

13.8 Convergence Analysis

To study convergence, we tracked the L^1 -norm of the difference between successive PageRank vectors across iterations.

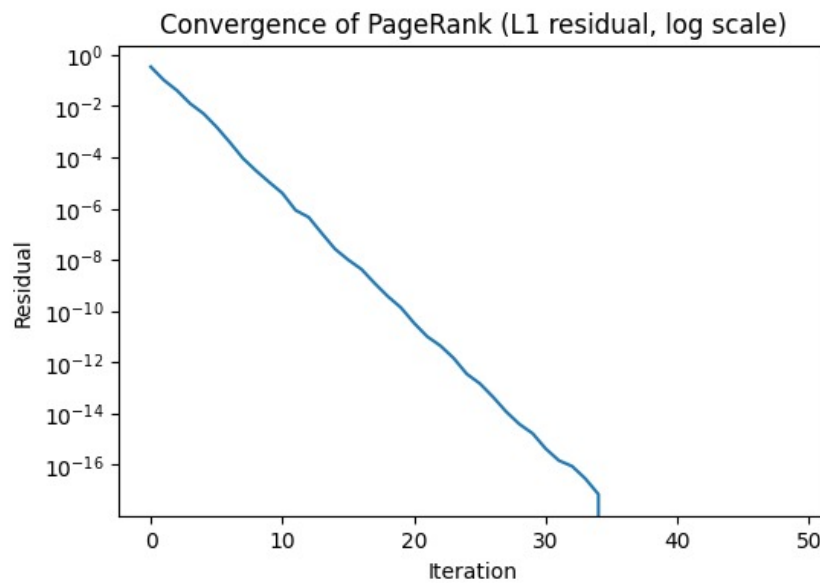


Figure 7: Convergence of PageRank

The convergence analysis was performed on the structured graph generated using preferential attachment (Barabási–Albert–type graph).

The residual decreases rapidly in the initial iterations and continues to decline steadily on a logarithmic scale. This indicates that the algorithm converges quickly in practice.

In our experiments, convergence was achieved within a few tens of iterations, confirming the efficiency of the implementation.

13.9 Graph Representations using PageRank

Beyond ranking nodes, the computed PageRank vectors can be used as feature representations for both nodes and entire graphs.

13.9.1 Node-Level Representation

Each node is assigned a PageRank score, which can be interpreted as a feature. This provides a simple one-dimensional embedding of nodes, where higher values correspond to more central or influential nodes.

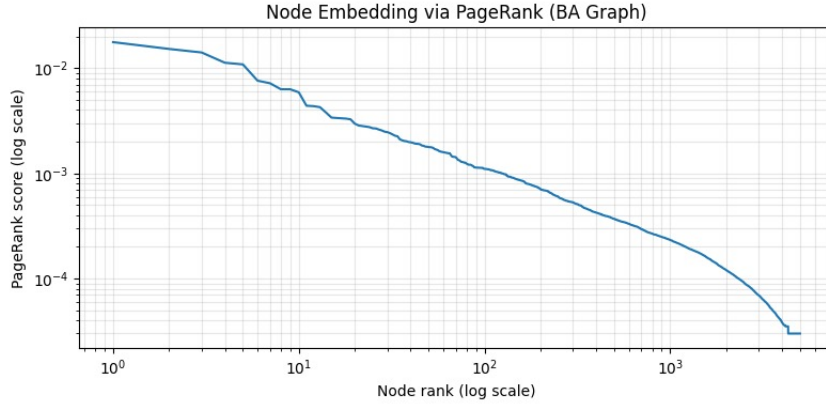


Figure 8: Node embedding rank plot

The sorted PageRank values provide a one-dimensional embedding of nodes. The steep decay indicates that a small number of nodes occupy highly central positions, while the majority of nodes have significantly lower importance. This reflects the hierarchical structure of the graph and highlights the disparity in node influence.

13.9.2 Graph-Level Representation

For each graph, the PageRank vector can be treated as a representation of the graph itself. Different graph structures produce distinct PageRank distributions.

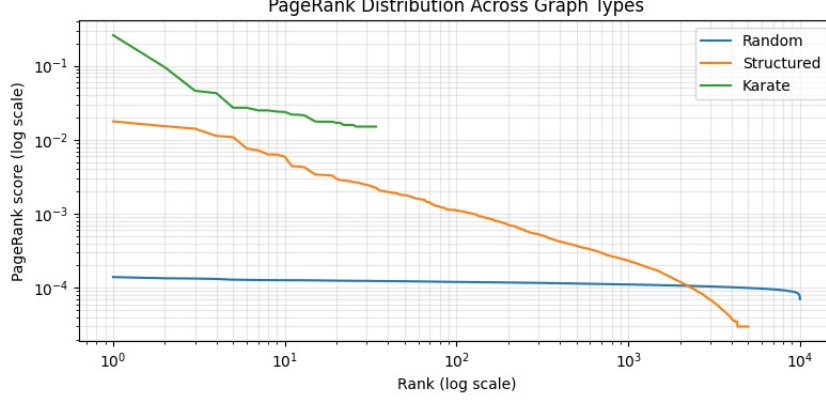


Figure 9: Rank plot comparison

The rank plot shows the distribution of PageRank scores across different graph types. The structured graph exhibits a clear heavy-tailed distribution, where a small number of nodes dominate in importance. In contrast, the random graph shows a nearly flat profile, indicating a more uniform distribution of importance. The Karate graph lies between these two extremes, reflecting moderate structural heterogeneity.

This demonstrates that PageRank captures global structural differences between graphs, and that the resulting distributions can be used as graph-level representations.

13.10 Connection to Graph Kernels

Beyond scalar importance scores, PageRank and Personalized PageRank can be used to construct feature representations.

In particular, PPR vectors encode how probability mass diffuses from a given node across the graph. These vectors can be treated as embeddings, where nodes with similar diffusion behavior have similar representations.

By aggregating such representations, one can compare graphs using similarity measures. This provides a connection between PageRank-based diffusion and graph kernel methods, where similarity is defined through feature representations derived from the graph structure.

13.10.1 Weisfeiler–Lehman Graph Kernel

To define a principled and widely-used similarity measure between graphs, we employ the Weisfeiler–Lehman (WL) subtree kernel. This kernel is based on an iterative node relabeling procedure that captures increasingly rich local neighborhood structure.

WL Relabeling Procedure Given a graph $G = (V, E)$ with initial node labels $l^{(0)}(v)$ (for unlabeled graphs, all nodes can be assigned the same initial label), the WL procedure updates node labels iteratively.

At iteration h , the label of node v is updated as:

$$l^{(h)}(v) = \text{hash}\left(l^{(h-1)}(v), \{l^{(h-1)}(u) : u \in \mathcal{N}(v)\}\right),$$

where $\mathcal{N}(v)$ denotes the neighbors of v , and the multiset of neighbor labels is sorted before hashing.

This process aggregates structural information from progressively larger neighborhoods. After h iterations, each node label encodes the subtree structure rooted at that node up to depth h .

Feature Map At each iteration h , we count the occurrences of node labels across the graph. This yields a feature vector:

$$\phi_h(G) \in \mathbb{R}^{d_h},$$

where each coordinate corresponds to the frequency of a particular label.

The overall feature representation is obtained by concatenating across iterations:

$$\phi(G) = [\phi_0(G), \phi_1(G), \dots, \phi_H(G)].$$

Kernel Definition The Weisfeiler–Lehman kernel between two graphs G_1 and G_2 is defined as the inner product of their feature vectors:

$$K(G_1, G_2) = \langle \phi(G_1), \phi(G_2) \rangle.$$

Equivalently,

$$K(G_1, G_2) = \sum_{h=0}^H \langle \phi_h(G_1), \phi_h(G_2) \rangle.$$

In practice, we use the normalized kernel

$$\tilde{K}(G_1, G_2) = \frac{K(G_1, G_2)}{\sqrt{K(G_1, G_1)K(G_2, G_2)}}$$

to ensure comparability across graphs of different sizes.

The WL kernel measures similarity between graphs based on the number of matching subtree patterns up to height H . Unlike simple degree-based or PageRank-based summaries, it captures higher-order structural information in a principled way.

We compute the similarity between the three graphs using the normalized WL kernel and present the resulting similarity matrix below.

	ER	BA	Karate
ER	1.0000	0.7668	0.6256
BA	0.7668	1.0000	0.4954
Karate	0.6256	0.4954	1.0000

Table 1: Weisfeiler–Lehman Kernel Similarity between Graphs

The normalized Weisfeiler–Lehman kernel matrix provides a comparison of the three graphs based on their local subtree structures. The ER and BA graphs exhibit the highest similarity (0.7668), indicating that despite their different generative mechanisms, they share many common local neighborhood patterns, particularly among low- and moderate-degree nodes. Since the WL kernel emphasizes local structure, these shared patterns lead to a relatively high similarity score.

The ER and Karate graphs have a moderate similarity (0.6256). While the Karate graph represents a real-world social network with some underlying organization, it still

contains many nodes with simple local neighborhoods. These overlap with patterns present in the ER graph, resulting in a non-negligible similarity.

The BA and Karate graphs show the lowest similarity (0.4954). This reflects the structural differences between the two: the BA graph is strongly dominated by hubs due to preferential attachment, whereas the Karate graph exhibits community structure and a more balanced distribution of node roles. As a result, their local subtree patterns differ more significantly.

Overall, these results highlight an important property of the WL kernel: it measures similarity through local neighborhood patterns rather than global topology. Consequently, graphs that differ in their global organization may still appear similar if they share common local structures, while graphs with distinct local patterns—such as the hub-dominated BA graph and the community-based Karate graph—exhibit lower similarity.

Although it may appear counterintuitive that the ER graph is more similar to the Karate graph than the BA graph, this is explained by the fact that the WL kernel measures similarity based on local neighborhood patterns rather than global structure. As a result, graphs with moderate and overlapping local structures can appear more similar than graphs with highly distinctive features such as strong hub dominance.

Properties

- It is invariant to node permutations (graph isomorphism invariant).
- It is well-defined for graphs of different sizes.
- It captures increasingly complex local structures with higher H .
- It is computationally efficient, with linear complexity in the number of edges per iteration.

A limitation of the WL kernel is that it primarily captures local structural patterns and may not fully distinguish graphs based on global topology.

13.10.2 Observations

From these representations, we observe:

- PageRank naturally provides node embeddings without additional learning.
- Different graph types produce distinct embedding distributions.
- Simple similarity measures can be constructed using these representations.

This demonstrates how PageRank connects graph structure to feature-based representations used in machine learning.

13.11 Summary

From the experiments, we observe the following:

- PageRank produces highly uneven distributions of importance across nodes.

- A small number of nodes dominate due to their structural position.
- The algorithm successfully identifies influential nodes in real-world networks.
- Personalized PageRank provides a localized view of influence.
- Sparse implementation ensures efficient computation and fast convergence.

13.12 Repository

The implementation for this experiment is available at:

<https://github.com/Arnab-Chakraborti/LAA-PageRank>

References

- [BP] Sergey Brin and Larry Page. *The Google Search Engine*. <http://google.stanford.edu>.
- [BGS05] Monica Bianchini, Marco Gori, and Franco Scarselli. Inside PageRank. *ACM Transactions on Internet Technology (TOIT)*, 5(1): 92–128, 2005.
- [CGMP98] Junghoo Cho, Hector Garcia-Molina, and Lawrence Page. Efficient crawling through URL ordering. In *Proceedings of the Seventh International World Wide Web Conference (WWW 98)*, 1998.
- [Gar95] Eugene Garfield. New international professional society signals the maturing of scientometrics and informetrics. *The Scientist*, 9(16), Aug 1995. http://www.the-scientist.library.upenn.edu/yr1995/august/issi_950821.html.
- [Gof71] William Goffman. A Mathematical Method for Analyzing the Growth of a Scientific Discipline. *Journal of the ACM*, 18(2), 1971.
- [Hav02] Taher Haveliwala. Topic-Sensitive PageRank. In *Proceedings of the 11th International Conference on World Wide Web (WWW '02)*, 2002.
- [Kle98] Jon M. Kleinberg. Authoritative sources in a hyperlinked environment. In *Proceedings of the 9th ACM-SIAM Symposium on Discrete Algorithms*, 1998.
- [New10] Mark Newman. *Networks: An Introduction*. Oxford University Press, 2010.
- [PBMW99] Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. *The PageRank Citation Ranking: Bringing Order to the Web*. Stanford InfoLab Technical Report, 1999.
- [SS11] Nino Shervashidze, Pascal Schweitzer, Erik Jan van Leeuwen, Kurt Mehlhorn, and Karsten M. Borgwardt. Weisfeiler-Lehman Graph Kernels. *Journal of Machine Learning Research*, 12: 2539–2561, 2011.
- [SS02] Bernhard Schölkopf and Alexander J. Smola. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press, 2002.
- [VSRB10] S. V. N. Vishwanathan, Nicol N. Schraudolph, Risi Kondor, and Karsten M. Borgwardt. Graph Kernels. *Journal of Machine Learning Research*, 11: 1201–1242, 2010.